

# **Telco Churn Prediction Project Steps:**

**Part 1: Project Introduction, Import Libraries, Load Dataset, Data Overview**

**Part 2: Data Cleaning and Preprocessing**

**Part 3: Exploratory Data Analysis (EDA)**

**Part 4: Feature Engineering and Data Preparation**

**Part 5: Baseline Machine Learning Models**

**Part 6: Hyperparameter Tuning**

**Part 7: SHAP Explainability**

**Part 8: Business Recommendations and Conclusion**

**Notebook now covers the complete lifecycle of a professional credit risk modeling project:**

notebook now follows a professional end-to-end workflow:

1. Project Introduction
2. Import Libraries
3. Load Dataset
4. Data Overview
5. Data Cleaning
6. Exploratory Data Analysis
7. Missing Value Treatment
8. Outlier Analysis
9. Feature Engineering
10. Feature Encoding
11. Feature Scaling
12. Train-Test Split
13. Logistic Regression
14. Random Forest
15. XGBoost
16. Baseline Model Comparison
17. Hyperparameter Tuning

18. Tuned Model Comparison
19. **SHAP Explainability**
20. **Business Recommendations**
21. **Project Conclusion**